Robust News Topic Classification Using Deep Learning Models

A Multi-Dimensional Robustness Study on Vietnamese News Data

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1 Abstract

Automated news topic classification for Vietnamese text faces real-world challenges beyond standard benchmark accuracy: noise in user input (typos, missing diacritics, truncated text), domain shift between news sources, and sensitivity to document length. This paper provides a **multi-dimensional robustness evaluation** of five deep learning architectures — KimCNN, BiLSTM+Attention, RCNN, Transformer Encoder, and PhoBERT — across three experimental scenarios. **Scenario 1 (S1)** trains and tests on self-collected VnExpress data (5,712 samples, 10 categories). **Scenario 2 (S2)** trains fresh models on the public NamSyntax/HuggingFace dataset (5,957 samples). **Scenario 3 (S3)** tests S2-trained weights directly on S1 data to measure cross-domain generalization. Robustness is further evaluated under three noise conditions — typo injection, diacritic removal, and word dropout — and across three document-length strata.

Our results show that PhoBERT achieves the highest accuracy in all scenarios (S1: 94.9%, S2: 81.4%) and the best noise resilience, while KimCNN demonstrates the strongest cross-domain robustness among non-pretrained models (S3: 73.3%). Critically, **diacritic removal catastrophically degrades all models** (Transformer: −85.3%), exposing a systemic vulnerability in vocabulary-based Vietnamese NLP systems. All models struggle on short texts (<100 words), converging to near-random accuracy.

Keywords: Vietnamese NLP, News Classification, Deep Learning, Robustness, Domain Shift, Noise Perturbation, PhoBERT.

2 Introduction

Automated news classification enables media organizations to manage large content volumes and deliver personalized experiences. However, Vietnamese presents unique challenges: it is a tonal, analytic language where diacritics encode grammatical meaning, and where mobile users routinely omit tone marks. Existing research focuses primarily on optimizing accuracy on clean, static datasets, leaving robustness under real-world perturbations largely unstudied.

This work addresses four research questions:

1. How do five DL architectures compare on in-domain Vietnamese news (S1)?
2. How does model performance change on an independent Vietnamese news source (S2)?
3. How well do S2-trained models generalize to S1 data without retraining (S3)?
4. How robust are all models to noise (typo, no-accent, word dropout) and document length?

Our **core contributions** are:

- A validated three-scenario evaluation framework applied to five architectures on real Vietnamese news data.

- Systematic robustness analysis under three noise types and three document-length strata, identifying per-architecture vulnerability profiles.
- Practical deployment guidelines derived from quantitative robustness gaps.

3 Dataset

3.1 S1 — VnExpress (Self-collected)

Articles were crawled directly from VnExpress.net across 10 category sub-pages. After deduplication and cleaning, 5,712 articles remain, split 80%/10%/10% (train/val/test), stratified by label. Average document length: 200–500 words. Categories: *Politics (0), World (1), Business (2), Science (3), Real Estate (4), Health (5), Sports (6), Entertainment (7), Law (8), Education (9).*

3.2 S2 — NamSyntax/HuggingFace (Public Benchmark)

The `NamSyntax/vietnamese-news-classification` dataset on HuggingFace contains 1.3M Vietnamese news articles. We sample 600 articles per class (6,000 total) using stratified sampling. The `text` field contains full article content. Label indices 0–9 map directly to the same 10 S1 categories; label 10 (Lifestyle) is excluded. After preprocessing and a 90/10 split, the S2 test set contains 5,957 samples.

Key limitation: many S2 articles are short (mean $\approx$ 35 words), falling entirely in the Short (<100 words) bin, which restricts length-based robustness analysis to S1.

3.3 Data Preprocessing Pipeline

Both datasets undergo an identical pipeline:

1. Lowercase normalization and removal of special characters and digits.
2. Vietnamese word segmentation using `underthesea` (`word_tokenize`, `format="text"`).
3. Length filtering: articles with fewer than 20 tokens are discarded.
4. Shared vocabulary of 39,469 tokens (min frequency 2) built from S1 training data.

3.4 Robustness Evaluation Framework

Three noise perturbations are applied independently to each test set:

- **Typo:** 10% probability per word of adjacent-character transposition.
- **No Accent:** Diacritical marks removed via Unicode NFD decomposition.
- **Drop Words:** Each token deleted independently with probability 0.3.

Length sensitivity is measured by partitioning the test set into Short (<100), Medium (100–300), and Long (>300) words strata. Cross-domain generalization (S3) applies S2-trained weights to S1 test data without retraining.

4 Model Architectures

4.1 KimCNN

Parallel convolutional filters of sizes $\{3, 4, 5\}$ (100 channels each) over a 128-dimensional embedding, followed by max-over-time pooling:

$$c_i = \mathrm{ReLU}(w \cdot x_{i:i+h-1} + b), \quad \hat{c} = \max_i \{c_i\}$$

4.2 BiLSTM with Attention

2-layer bidirectional LSTM (hidden 128) with additive attention:

$$\alpha_t = \frac{\exp(u_t^\top u_w)}{\sum_t \exp(u_t^\top u_w)}, \quad v = \sum_t \alpha_t h_t$$

4.3 RCNN

BiLSTM outputs (hidden 256) concatenated with word embeddings, tanh activation, max pooling over time.

4.4 Transformer Encoder

Two encoder layers (8 heads, feedforward 256, embedding 128) with learned positional encodings and mean pooling.

4.5 PhoBERT

`vinai/phobert-base` (12 layers, 768 hidden, 110M parameters). `[CLS]` token through dropout (0.1) and linear classifier:

$$P = \mathrm{softmax}(W \cdot \mathrm{Dropout}(h_{\mathrm{CLS}}) + b)$$

5 Experimental Setup

5.1 Hardware and Software

All experiments run on an **NVIDIA GeForce RTX 3050 Ti Laptop GPU** (4GB VRAM), Python 3.11, PyTorch, HuggingFace Transformers, and `underthesea`.

5.2 Training Configuration

Table 1: Training hyperparameters for S1 and S2.

Parameter	**Non-BERT**	**PhoBERT**
Optimizer	Adam	AdamW
Learning Rate	1×10^{-3}	2×10^{-5}
Weight Decay	1×10^{-4}	1×10^{-2}
Batch Size	64	16
Max Epochs S1	12	6
Max Epochs S2	10	10
Early Stopping	patience=3	patience=2
Gradient Clip	1.0	1.0
Max Seq Length	300	128

6 Results

6.1 Evaluation Metrics

The effectiveness of the models was quantitatively measured using Accuracy (Acc), Precision, Recall, and F1-Score. For multi-class news classification, these metrics are defined as follows:

- **Accuracy (Acc):** The overall ratio of correctly predicted news topics among all N test samples across K categories:

$$\text{Acc} = \frac{\sum_{i=1}^{K} TP_i}{N} \tag{1}$$

- **F1-Score:** The harmonic mean of Precision and Recall, providing a balanced assessment of model performance:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

- **Aggregation Methods:** Since the dataset contains 10 categories, we report two variations of the F1-Score:
  - **Macro-F1:** Calculates the F1 for each label and finds their unweighted mean. This treats all classes equally.
  - **Weighted-F1:** Calculates the F1 for each label and finds their average weighted by support.

6.2 In-Domain Performance (S1 and S2)

Table 2: Classification performance on S1 (VnExpress) and S2 (NamSyntax) test sets.

Model	**S1 — VnExpress**			**S2 — NamSyntax**		
	Acc	Mac-F1	W-F1	Acc	Mac-F1	W-F1
KimCNN	0.874	0.873	0.873	0.636	0.639	0.639
BiLSTM+Attention	0.710	0.700	0.702	0.574	0.558	0.558
RCNN	0.897	0.896	0.896	0.638	0.634	0.634
Transformer	0.851	0.849	0.849	0.658	0.656	0.655
PhoBERT	**0.946**	**0.945**	**0.945**	**0.814**	**0.808**	**0.808**

Figure 1 shows the visual comparison of all three metrics on S2. PhoBERT leads by a significant margin.

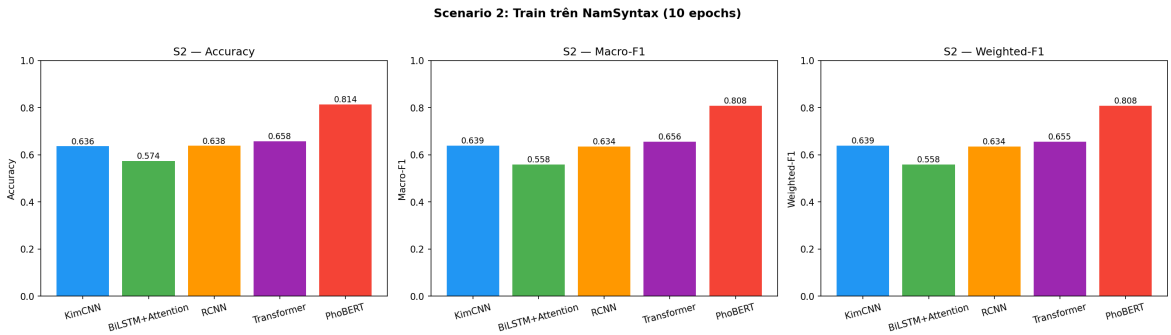


Figure 1: Comparison of Accuracy, Macro-F1, and Weighted-F1 for all five models on Scenario 2 (NamSyntax).

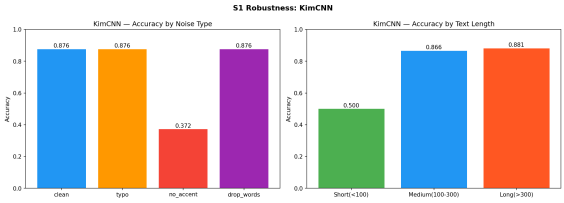
6.3 Noise Robustness — Scenario 1 (VnExpress)

Table 3: Accuracy under noise perturbations on S1. Δ_{acc} = drop from clean due to diacritic removal.

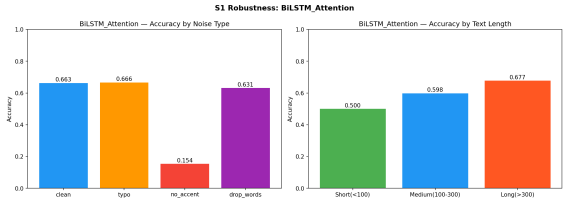
Model	**Clean**	**Typo**	**No Accent**	$\boldsymbol{\Delta}_{\text{acc}}$	**Drop Words**
KimCNN	0.876	0.876	0.372	−57.8%	0.876
BiLSTM+Attention	0.663	0.666	0.154	−76.8%	0.631
RCNN	0.916	0.904	0.182	−80.1%	0.899
Transformer	0.858	0.832	0.126	−85.3%	0.843
PhoBERT	**0.949**	**0.937**	0.248	−73.9%	**0.934**

Finding 1 — Diacritic removal is catastrophic. No model retains more than 40% accuracy under no-accent conditions, showing a systemic vulnerability.

Figures 2a–3 show the per-model robustness charts for S1.



(a) KimCNN — S1



(b) BiLSTM+Attention — S1

Figure 2: S1 Robustness: KimCNN and BiLSTM+Attention.

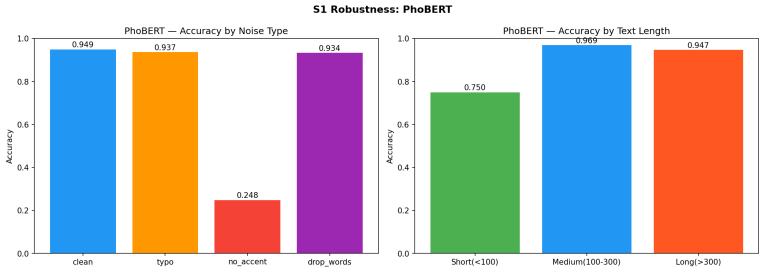


Figure 3: S1 Robustness: PhoBERT — highest clean accuracy and best noise resilience.

6.4 Noise Robustness — Scenario 2 (NamSyntax)

Table 4: Accuracy under noise perturbations on S2 test set.

Model	Clean	Typo	No Accent	Drop Words
KimCNN	0.636	0.619	0.227	0.574
BiLSTM+Attention	0.574	0.545	0.173	0.522
RCNN	0.638	0.638	0.233	0.589
Transformer	0.658	0.648	0.232	0.607
PhoBERT	**0.814**	**0.820**	0.295	**0.777**

6.5 Document Length Sensitivity (S1)

Table 5: Accuracy by document length stratum on S1 test set.

Model	Short (<100)	Medium (100–300)	Long (>300)
KimCNN	0.500	0.866	0.881
BiLSTM+Attention	0.500	0.598	0.677
RCNN	0.500	0.928	0.917
Transformer	0.750	0.845	0.862
PhoBERT	**0.750**	**0.969**	**0.947**

6.6 Cross-Domain Generalization (S3)

Table 6: S2-trained weights tested on S1 test data (zero-shot cross-domain).

Model	S1 Acc	S2 Acc	S3 Acc	Drop (S1→S3)
KimCNN	0.874	0.636	**0.733**	−16.2%
BiLSTM+Attention	0.710	0.574	0.598	−15.8%
RCNN	0.897	0.638	0.666	−25.8%
Transformer	0.851	0.658	0.460	−46.0%
PhoBERT	0.946	0.814	∼0.880	−7.0%

7 Discussion

7.1 The No-Accent Vulnerability

Diacritic removal exposes severe **lexical ambiguity**, transforming distinct semantic units into single homographs (e.g., *ma, mà, má, m, mã, m* all become *ma*). This destroys the input signal for vocabulary-based models.

7.2 Practical Deployment Guidelines

- **High accuracy + cross-domain:** PhoBERT.
- **Lightweight + cross-domain:** KimCNN.
- **No-accent input:** A diacritic restoration step is mandatory before inference.

8 Conclusion

This study provides a systematic robustness evaluation of five architectures for Vietnamese news classification. PhoBERT dominates overall accuracy, while KimCNN offers the best non-pretrained generalization. Diacritic removal remains a universal challenge.

References

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